

CENTRAL CITY GENERAL HOSPITAL

Department of Pulmonology & Infectious Diseases

Date of Consultation: September 16, 2026
Attending Physician: Dr. Sarah Jenkins, MD

Patient ID: MRN-847201

PATIENT DEMOGRAPHICS

Name:	Alex Mercer	Age/DOB:	45 Y (DOB: 04/12/1981)
Gender:	Male	Height/Weight:	178 cm / 64 kg

CLINICAL HISTORY

Chief Complaint: The patient presents with a persistent, productive cough lasting for the past 4 weeks, accompanied by yellowish sputum.

History of Presenting Illness (HPI): Mr. Mercer reports experiencing profound fatigue, recurrent low-grade fevers (typically peaking in the late afternoon), and severe drenching night sweats. He has experienced a documented unintentional weight loss of approximately 6 kg over the last 8 weeks. He denies any recent travel or known exposure to individuals with active TB, though he works in a high-density occupational setting.

PHYSICAL EXAMINATION

- Vitals:** BP: 118/76 mmHg | HR: 88 bpm | Temp: 38.2°C (100.8°F) | SpO2: 96% on room air.
- General:** Patient appears fatigued and mildly cachectic. No cyanosis or clubbing observed.
- Respiratory:** Auscultation reveals decreased breath sounds and bronchial breathing over the right upper lung field. Post-tussive crepitations (crackles) are noted in the right apical region.

DIAGNOSTIC INVESTIGATIONS

1. Radiological Findings:

Chest X-Ray (PA View): Reveals bilateral patchy consolidations with a distinct 3 cm cavitary lesion in the right upper lobe. No significant pleural effusion noted.

2. Microbiological and Molecular Pathology:

Sputum Acid-Fast Bacillus (AFB) Smear: Positive (3+). Indicates a high bacterial load.

GeneXpert MTB/RIF Assay:

- Mycobacterium tuberculosis* complex: **DETECTED**
- Rifampicin Resistance: **NOT DETECTED**

Complete Blood Count (CBC): Mild normocytic anemia; elevated Erythrocyte Sedimentation Rate (ESR) at 72 mm/hr.

ASSESSMENT & DIAGNOSIS

Primary Diagnosis: Active Pulmonary Tuberculosis (Drug-Susceptible)

TREATMENT PLAN & RECOMMENDATIONS

1. **Medication (Intensive Phase):** Initiate first-line Anti-Tubercular Therapy (ATT) immediately.
 - Isoniazid (H) 300 mg daily
 - Rifampicin (R) 600 mg daily
 - Pyrazinamide (Z) 1500 mg daily
 - Ethambutol (E) 1200 mg daily
 - *Pyridoxine (Vitamin B6) 25 mg daily to prevent isoniazid-induced neuropathy.*
2. **Infection Control:** Patient is advised to strictly isolate at home for the next 2-3 weeks until sputum smears become negative. Must wear a high-quality (N95) mask when in the presence of household members.
3. **Follow-up:**
 - Liver function tests (LFTs) scheduled in 2 weeks to monitor for potential medication toxicity.
 - Repeat sputum AFB smear in 4 weeks.

Dr. Sarah Jenkins, MD
Pulmonologist, Central City General Hospital
Electronically Signed: Sept 16, 2026